University of Milano - Bicocca

School of Science - Department of Informatics, Systems and Communication

Master's Degree Program in Data Science

Data Management Project

Environmental and Seasonal Effects on Football Match Performance: Analysis of the Big Five European Leagues over the Last Three Seasons

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Abstract

This project investigates the relationship between environmental factors and football match performance, with a particular focus on weather conditions and seasonal effects. Match-level data from the top five European leagues over the last three seasons were collected through web scraping and subsequently enriched with historical meteorological information. The resulting data were integrated into a unified relational database containing match statistics, stadium information, team attributes, temporal features, and weather conditions.

This structured database enables efficient querying and supports comprehensive analysis aimed at addressing the proposed research questions. To ensure data quality and reliability, completeness and consistency were systematically assessed across all variables, with particular attention paid to the identification and treatment of missing values. Overall, this work provides a data-driven framework for exploring how meteorological and temporal factors may influence soccer match outcomes and performance indicators.

Contents

1 Introduction

Over the years, soccer has become one of the most popular sports worldwide and has undergone significant evolution from both a technical and a tactical perspective. In parallel, the increasing availability of structured data and the widespread adoption of technological tools have profoundly changed how matches are analysed. Modern data-driven approaches support coaching staff, analysts, and clubs in evaluating performance, identifying patterns, and improving decision-making processes. Despite the growing abundance of match-related data, the influence of external environmental factors, such as weather conditions and seasonality, remains relatively underexplored. Most existing analysis primarily focus on technical and tactical indicators, while meteorological and temporal variables are often treated as secondary or analysed independently. However, weather conditions and seasonal effects can plausibly influence match dynamics, physical intensity, stadium attendance, and overall team performance.

1.1 Objective and Research Questions

The goal of this project is to investigate the relationship between weather conditions, seasonal effects, and soccer match performance by integrating match statistics with meteorological and seasonal data. In particular, match-level data were collected from FBref and enriched with weather information retrieved through the Open-Meteo Historical Weather API, as well as temporal features capturing seasonality. This integrated dataset enables a more comprehensive analysis of how environmental and temporal factors may influence match outcomes and in-game events, with potential applications in performance analysis and coaching strategies.

While football match outcomes are generally influenced by a wide range of factors, including team quality, tactical choices, physical conditions, and contextual variables, this project concentrates on the environmental and temporal aspects that are less frequently addressed in the literature. Based on the available data, we formulated the following research questions:

- Does weather influence match outcomes and match statistics?
- Do seasonal effects influence match outcomes and match statistics?

1.2 Structure

The remainder of this report is structured as follows. Chapter 2 describes the data acquisition and enrichment process, detailing the sources used and the methods adopted to collect and integrate additional information. Chapter 3 presents the data storage layer, including the database design and the adopted relational model. Chapter 4 discusses the research questions and analytical approach, focusing on visualizations and statistical tests. Chapter 5 is dedicated to data quality, addressing the dimensions of completeness and consistency, and describing the strategies adopted to handle detected issues. Finally, Chapter 6 concludes the work by summarizing the main findings and highlighting possible future developments.

1.3 Software Architecture

The overall project follows a modular, pipeline-based architecture designed to ensure maintainability and reproducibility. The system can be conceptually divided into four main layers: data acquisition and enrichment, data storage, analysis and visualization, and data quality. This layered organization provides a clear framework for both implementation and discussion.

From an architectural perspective, the system is conceptually organized as a layered data platform where each layer implements dedicated Python modules and diverse technologies. The data acquisition and enrichment layer is responsible for collecting raw data through web scraping and integrating additional information retrieved through external APIs. All data acquisition and enrichment tasks are implemented in Python (3.12.7) and divided into several steps. Then, the data storage layer provides a structured representation of the collected data, leveraging PostgreSQL (18.1) with the PostGIS extension as the Relational Database Management System (RDBMS), or more specifically, as the Object-Relational Database Management System (ORDBMS). The analysis and visualization layer uses the stored data to address the research questions of the project. This layer includes exploratory analysis, statistical tests, and visualizations, all implemented using Python-based tools. Finally, the data quality layer focuses on assessing and improving the reliability of the dataset. Validation checks and corrective strategies are applied to identify and handle potential data quality issues.

Considering the architecture described above, the Jupyter Notebook is structured as a modular workflow that calls functions and classes defined in dedicated Python modules, corresponding to the different layers of the overall workflow. This structure facilitates code reuse and reproducibility of the pipeline, while also improving readability. Additionally, Git was adopted as a version control system to support collaborative development.

2 Data Acquisition and Enrichment

This chapter describes the data acquisition and enrichment phases of the project. Data collection was performed using two complementary approaches: web scraping and API-based data retrieval. In particular, match statistics were obtained through web scraping from FBref, while external APIs, such as the OpenCage Geocoding API and the Open-Meteo Historical Weather API, were employed to enrich the dataset with geographical and meteorological information. The aforementioned data sources were chosen based on reliability, coverage, and relevance to the research questions.

2.1 Web Scraping for Acquisition

Web scraping is the automated process of extracting useful information from web pages, when data are not directly available via official APIs or downloadable datasets. Historically, most websites provide their content through HTML documents, which can be parsed to retrieve relevant elements, such as text, tables, and links. However, modern websites often employ dynamic content loading, client-side rendering, or protection mechanisms against automated requests. As a result, simple HTTP-based scraping approaches may be insufficient. In such cases, browser automation tools are required to simulate real user behaviour and correctly retrieve the rendered content while avoiding access restrictions.

2.1.1 Scraping from FBref

FBref (`https://fbref.com/en`) is a web-based platform that provides comprehensive football statistics for teams and players. It covers the top five European leagues (Premier League, La Liga, Serie A, Bundesliga, and Ligue 1), as well as international competitions and historical seasons. A key reason for selecting FBref as the primary data source is the reliability of its data, which is largely supplied by Opta, a well-established sports analytics provider. Moreover, FBref offers detailed match-level information, including final results, expected goals (xG), match dates, stadium details, attendance figures, disciplinary events, and a wide range of performance metrics. Its richness and granularity make FBref particularly suitable for analytical and data-driven studies.

Match statistics were collected using a custom web scraper implemented in Python. From a structural perspective, FBref pages are based on standard HTML, which makes them suitable for parsing with libraries such as `Beautiful Soup`. In principle, this would allow for data extraction using simple HTTP requests. However, during the development of the scraper, direct requests always triggered HTTP 403 Forbidden errors, indicating that automated access was being blocked.

To overcome this limitation, the Python library `Selenium` was employed to simulate a real web browser and the corresponding user interaction. By controlling a browser driver, `Selenium` allows the scraper to load pages in the same way a human user would, effectively bypassing access restrictions and ensuring that all content is correctly rendered, while also providing robustness against potential changes in client-side behaviour. Once the page was fully loaded by `Selenium`, `Beautiful Soup` was then used to parse the HTML source and extract the relevant data.

The scraper followed an automatic hierarchical navigation strategy, collecting match-level data for the big five European leagues across the last three completed seasons (2022/2023, 2023/2024, 2024/2025). To achieve this objective, the scraper started from the competition-level page (`https://fbref.com/en/comps`), extracted the top five leagues table, iterated over the selected seasons, and accessed the corresponding Scores and Fixtures page for each of them. From this page, links to individual match report pages were extracted. Each match report page contains detailed statistics for both teams, which were parsed and stored. This process was repeated for every selected competition. In particular, the following information was extracted:

- Team names.
- Match date and kick-off time.
- Final score and expected goals (xG).
- Stadium name, city, and attendance.
- Ball possession.
- Passes (completed and attempted).
- Shots (on target and total).
- Saves (made and faced).
- Disciplinary events (yellow and red cards).
- Additional performance metrics, such as fouls, corners, crosses, touches, tackles, interceptions, aerial duels, clearances, offsides, goal kicks, throw-ins, and long balls.

Due to the volume of data collected, the scraping process required several hours to complete. To avoid repeated extractions, all retrieved data were temporarily stored in a CSV file. In total, the scraper processed 5,330 matches, distributed as per Table 2.1.

	2022/2023	**2023/2024**	**2024/2025**
Premier League	380	380	380
La Liga	380	380	380
Ligue 1	306	306	380
Bundesliga	306	306	306
Serie A	380	380	380

Table 2.1: Number of matches per competition and season

The number of matches reported in the table is not arbitrary, as it reflects the official structure of each league. Differences across competitions and seasons arise from the varying number of participating teams and the corresponding league formats. In particular, leagues composed of 20 teams feature 380 matches per season, while leagues with 18 teams consist of 306 matches. The variation observed for Ligue 1 is due to the change in the number of teams across the considered seasons.

2.2 Data Cleaning and Schema Restructuring

After the scraping phase, the collected dataset underwent a data cleaning and restructuring process aimed at handling missing values and preparing the data for storage in a relational database.

First, the presence of missing values was systematically evaluated across all columns. Overall, no critical issues were identified; however, a small number of missing values required specific attention. In particular, missing values were observed in save-related statistics. On FBref, when no saves occur during a match, the corresponding row is omitted from the statistics table rather than being reported as 0. As a result, missing values in these fields do not indicate data loss but reflect the absence of recorded events. To handle these cases, matches in which both teams registered 0 shots on target were identified. In these five instances, the associated save-related metrics were set to 0, as no saves could have occurred by definition. Attendance figures were also inspected. Seven matches contained missing attendance values. When possible, these values were manually retrieved from a reliable external source (`https://www.diretta.it/`), a widely used football statistics platform that provides official match reports. Attendance data were successfully recovered for five of these matches. All manual corrections were explicitly documented to ensure transparency and traceability. Further considerations regarding these and other potential data issues will be addressed in the context of the overall data quality assessment. Beyond the data cleaning process, the dataset was restructured to better align with a relational database schema. Separate tables were created for competitions, teams, stadiums, matches, and match statistics. This normalization process minimized redundancy and improved data integrity. Unique identifiers were assigned to each entity, and foreign keys were introduced to represent relationships between tables. After restructuring, the data were ordered by season, competition, and match date to ensure chronological consistency.

2.3 APIs and Data Enrichment

While web scraping provided the core match statistics, additional contextual and environmental information was required to address the research questions. This data was obtained through external APIs, which enabled the integration of geographical, temporal, and meteorological information not available on FBref. In this phase, the use of APIs allowed us not only to enrich the existing dataset but also to introduce new variables, expanding the analytical scope of the project.

2.3.1 Meteorological Season and Time Slot

First, temporal attributes were derived from the match date and kick-off time. Each match was assigned a `meteorological_season` (Summer, Autumn, Winter, or Spring) and a `time_slot`

(Midday, Afternoon, Evening, or Night). The time slots were defined according to match kick-off times as follows: Midday (11:00–13:59), Afternoon (14:00–16:59), Evening (17:00–19:59), and Night (20:00–22:59). These features allowed for the analysis of seasonal patterns on match outcomes and performance, while time-dependent effects were left for potential applications in extended studies.

2.3.2 Stadium Geocoding

Retrieving historical weather data requires geographical coordinates as input. Since the scraping process extracted only stadium names and cities, a geocoding step was necessary. Initially, the Nominatim API was tested due to its free access and lack of authentication requirements. However, this solution failed to resolve approximately 10% of the stadium locations, leading to incomplete spatial coverage. For this reason, the OpenCage Geocoding API was adopted as it provided more accurate and consistent results, successfully returning coordinates for all stadiums. In particular, a short delay between consecutive requests was introduced to comply with the API rate limits and ensure stable execution of the geocoding process. Table 2.2 summarizes the input parameters provided to the API, while Table 2.3 reports the output fields extracted and stored in the dataset.

Input Parameter	**Description**
API key	Authentication key required to access the OpenCage Geocoding API
Stadium name	Name of the stadium extracted from FBref
City	City where the stadium is located extracted from FBref
Language	Response language set to English (`language = en`)
Result limit	Maximum number of returned results set to one (`limit = 1`)
Annotations	Disabled to reduce response size (`no_annotations = 1`)

Table 2.2: Input parameters for the OpenCage Geocode API

Output Field	**Description**
Latitude	Latitude of the stadium location
Longitude	Longitude of the stadium location
Country	Country where the stadium is located, extracted from the formatted address
Geometry	Geographical point representation of the stadium (longitude, latitude)

Table 2.3: Output values retrieved from the OpenCage Geocode API

2.3.3 Timezone Identification

Once geographical coordinates were obtained, the local `timezone` of each stadium was determined using the `timezonefinder` Python package. This step was required to correctly align match kick-off times with the corresponding weather data, ensuring temporal consistency between match events and meteorological observations.

2.3.4 Historical Weather Data

Historical weather data were retrieved using the Open-Meteo Historical Weather API, chosen for its free access for non-commercial use, high spatial resolution, and extensive historical coverage.

The API provides detailed information on variables such as precipitation, wind speed, and temperature, collected on an hourly or daily basis depending on project needs. Moreover, historical data are available for up to 80 years, offering very broad temporal coverage.

To account for weather variability before and during the match, we decided to collect data within a temporal window spanning from two hours before to two hours after the scheduled kick-off. For each variable, the average value over this interval was computed, providing a robust representation of the weather conditions experienced during the match. In particular, the input parameters supplied to the Open-Meteo API and the output values extracted and integrated into the dataset are summarized in Tables 2.4 and 2.5, respectively.

Input Parameter	**Description**
Latitude	Latitude of the stadium, obtained from OpenCage Geocoding API
Longitude	Longitude of the stadium, obtained from OpenCage Geocoding API
Start date	Start date of the data retrieval period (the day before the match)
End date	End date of the data retrieval period (the day after the match)
Hourly variables	List of meteorological variables to retrieve (`temperature_2m`, `precipitation`, `wind_speed_10m`)
Timezone	Local timezone of the stadium to align hourly data with the match time

Table 2.4: Input parameters for the Open-Meteo Historical Weather API

Output Field	**Description**
Precipitation	Average precipitation (mm) within the ±2-hour window
Wind Speed	Average wind speed (km/h) within the ±2-hour window
Temperature	Average temperature (°C) within the ±2-hour window

Table 2.5: Output values retrieved from the Open-Meteo Historical Weather API

2.3.5 Weather Categorization

In addition to storing continuous meteorological variables, weather conditions were categorized into discrete classes. This transformation was motivated by analytical considerations: categorical variables improve interpretability, facilitate comparisons across matches, and support grouped analysis and visualization. Specifically, the following categorizations were applied:

- `is_precipitation`: a boolean indicator reflecting whether any precipitation occurred during the ±2-hour match window (0=no, 1=yes).
- `wind_speed_category`: wind speed was classified into Low (≤7 km/h), Medium (8–28 km/h), and High (>28 km/h), based on a simplified Beaufort scale.
- `temperature_category`: temperatures were divided into Cold (≤12°C), Mild (13–25°C), and Hot (>25°C).

3 Data Storage

This chapter describes the data storage strategy adopted, including the choice of the data model and the database management system. Indeed, an appropriate data storage design is essential to ensure data integrity, efficient querying, and scalability throughout the analysis workflow. For this project, we chose to employ a relational data model, in which data is organized into tables, each representing a specific entity, and the relationships between them are explicitly defined through primary and foreign keys. Primary keys uniquely identify each record, while foreign keys enforce referential integrity across related tables, linking records together in a consistent manner.

We opted for the relational model for the following reasons:

- The project entities are well-defined and naturally map to relational tables.
- The explicit structure of tables and relationships promotes data consistency and makes the connections between different types of information clear and intuitive.
- The overall data volume is moderate, making a relational database well suited to the project requirements. In contrast, a document-based approach would have introduced unnecessary redundancy and data duplication.
- Some attributes, such as match results and performance statistics, may be subject to errors or inconsistencies. Thus, the relational model supports integrity constraints, like `NOT NULL`, `UNIQUE`, and `CHECK`, that help preserve data correctness.
- The use of complex queries involving `JOIN` and `GROUP BY` operations enables in-depth exploratory and analytical analysis.

Among the available database management systems, we chose PostgreSQL over more common alternatives such as MySQL due to its robustness, extensibility, and advanced feature set. PostgreSQL offers performance optimizations typically found in enterprise-grade systems and supports complex analytical workloads. In particular, through the PostGIS extension, it provides advanced geospatial functionalities and data types, such as `geometry(Point)`, which were used to store stadium locations and to enable efficient spatial queries. Additionally, PostgreSQL ensures high reliability and performance under concurrent access scenarios by adopting a Multi-Version Concurrency Control (MVCC) mechanism, which allows read and write operations to occur simultaneously without blocking. As a fully relational system, it also offers comprehensive SQL (Structured Query Language) support, enabling the use of advanced querying and analytical techniques.

3.1 Database Structure

The database consists of six tables connected by six relationships. Figure 3.1 provides an overview of the database structure, highlighting the main entities and their attributes. Additionally, this schema facilitates understanding of how information about competitions, teams, matches, stadiums, statistics, and weather conditions is organized and interconnected within the database.

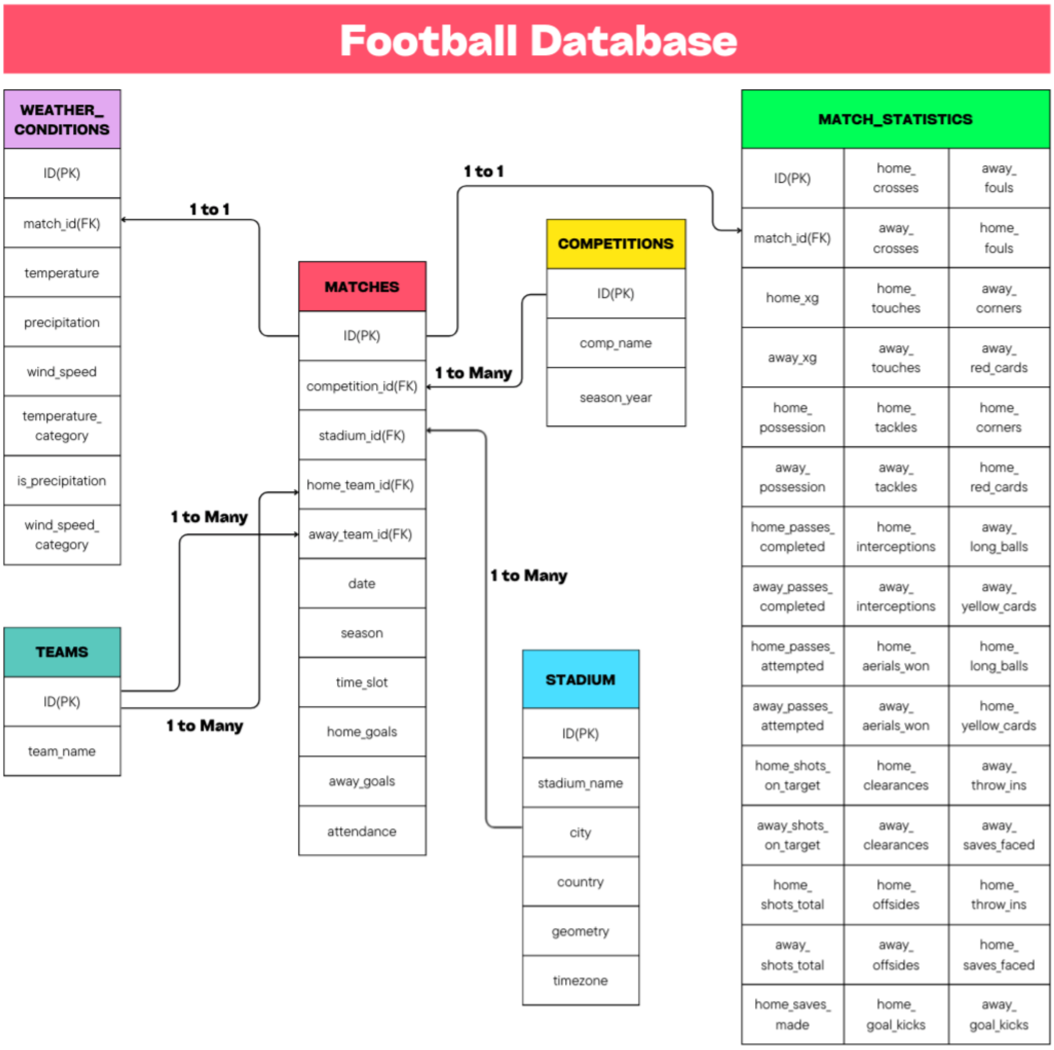


Figure 3.1: Database structure diagram

The main entities represented in the database are:

- `Matches`: stores information related to individual football matches, including the match date, team names, final result, associated competition, and attendance.
- `Competitions`: collects details about football competitions, such as the competition name and the corresponding season.

- `Stadiums`: represents the venues where matches are played, storing information such as the name, city, country, timezone, and geographical coordinates.
- `Match_Statistics`: contains performance metrics associated with each match, including expected goals, passes, shots, saves, cards, and other relevant statistics useful for analysis.
- `Weather_Conditions`: stores meteorological variables related to each match, such as precipitation, wind speed, and temperature, including both their values and categories.
- `Teams`: lists all football teams included in the database, identified by their names.

Moreover, the relationships between the tables are defined as follows:

- `Matches` to `Match_Statistics` (1 to 1): each match is associated with exactly one statistics record, and each set of statistics corresponds to one and only one match.
- `Matches` to `Weather_Conditions` (1 to 1): each match is linked to a single record describing the weather conditions during the game, and each weather record is associated with exactly one match.
- `Competitions` to `Matches` (1 to Many): a competition can include multiple matches, while each match belongs to exactly one competition.
- `Stadiums` to `Matches` (1 to Many): a stadium can host multiple matches over time, whereas each match is played in a single stadium.
- `Teams` (Home) to `Matches` (1 to Many): a team can play multiple matches as the home team, while each match has exactly one home team.
- `Teams` (Away) to `Matches` (1 to Many): a team can play multiple matches as the away team, while each match has exactly one away team.

3.2 Queries

SQL is the standard language for querying relational databases. It is a declarative language that allows users to retrieve and combine information from multiple tables efficiently. In our database, for instance, it is possible to join data on matches, competitions, stadiums, match statistics, weather conditions, and teams, obtaining a complete view of the available information.

As an example, consider the hypothetical scenario in which the San Siro stadium in Milan is unavailable for watching a match. As soccer fans, we may want to identify other stadiums within a 50-kilometre radius to find alternative match locations without straying too far from the city of Milan. In this case, a spatial query using the PostgreSQL `geometry` type, as shown in Figure 3.2, allows us to perform this analysis.

```
WITH san_siro AS (
    SELECT id, geometry::geography AS pos
    FROM stadiums
    WHERE stadium_name ILIKE '%San Siro%' OR stadium_name ILIKE '%Meazza%'
    LIMIT 1
)
SELECT
    s.stadium_name AS "Stadium Name",
    ROUND((ST_Distance(s.geometry::geography, ss.pos) / 1000)::numeric, 2)
    AS "Distance (km)"
FROM stadiums s, san_siro ss
WHERE ST_DWithin(s.geometry::geography, ss.pos, 50000) AND s.id <> ss.id
ORDER BY "Distance (km)";
```

Figure 3.2: Query 1 code - finding stadiums within 50 km of San Siro

The output of this query, shown in Table 3.1, lists the stadiums located within a 50-kilometre radius of San Siro, including U-Power Stadium in Monza, Stadio Giuseppe Sinigaglia in Como, and Gewiss Stadium in Bergamo, along with their corresponding distances.

	Stadium Name	**Distance**
1	U-Power Stadium	18.51
2	Stadio Giuseppe Sinigaglia	37.52
3	Gewiss Stadium	48.82

Table 3.1: Query 1 result - stadiums within 50 km of San Siro

Another example of a query involves identifying the teams with the highest total number of home goals scored in the top five European leagues across the last three seasons. This analysis provides a direct measure of teams' offensive output during home matches. Figure 3.3 shows the SQL query.

```
SELECT t.team_name AS "Team Name", SUM(m.home_goals) AS "Total Home Goals"
FROM matches m
JOIN teams t ON m.home_team_id = t.id
GROUP BY team
ORDER BY total_goals DESC
LIMIT 10;
```

Figure 3.3: Query 2 code - finding top 10 teams by total home goals

The results, presented in Table 3.2, indicate that Bayern Munich is the team that scored the highest number of goals in home matches, followed by Manchester City and Liverpool. This data reflects actual offensive performance and provides a basis for further analysis, showing which teams are expected to offer an exciting show. Furthermore, most of the top-scoring teams belong to the Bundesliga, Premier League, and La Liga. No Serie A teams appear in the top 10, which may reflect differences in tactical strategies and styles of play commonly observed in this league, where matches tend to be more defensive and goal-scoring opportunities are more limited. In general, it is important to note that the query only considers goals scored in home matches and does not account for expected goals or other performance metrics.

	Team Name	**Total Home Goals**
1	Bayern Munich	159
2	Manchester City	154
3	Liverpool	137
4	Real Madrid	137
5	Dortmund	136
6	Arsenal	136
7	Paris S-G	132
8	Barcelona	132
9	Newcastle Utd	125
10	Atlético Madrid	125

Table 3.2: Query 2 result - top 10 teams by total home goals

Beyond satisfying user curiosity, we employed similar queries throughout the subsequent phases of the project to explore the data, support the analysis conducted to address the research questions, and evaluate data quality.

4 Exploratory Data Analysis

The goal of the exploratory data analysis (EDA) is to provide an initial understanding of the main characteristics of the data and to identify preliminary patterns related to the research questions. This phase combines descriptive statistics, tests, and visualizations to highlight meaningful associations and trends that may justify further investigation. In the first step, the analysis examined the relationship between weather conditions and match outcomes or match-related statistics. Subsequently, potential seasonal effects were also taken into account.

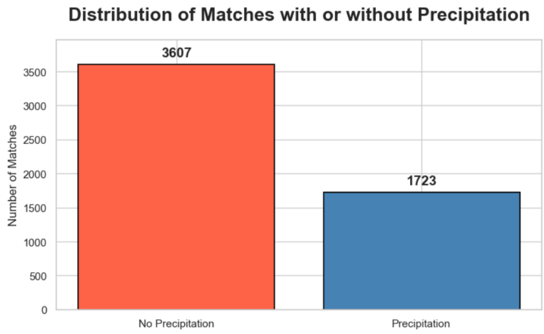
4.1 Does Weather influence Match Outcomes and Match Statistics?

This research question investigates whether meteorological factors, such as precipitation, wind speed, and temperature, influence football match results and the intensity of play. To address this question in a structured way, the analysis was divided into two main parts. First, the effect of weather conditions on match outcomes, in terms of home wins, draws, away wins, was examined. Then, the focus shifted to match statistics, providing a more detailed view of game intensity and playing style under different weather conditions.

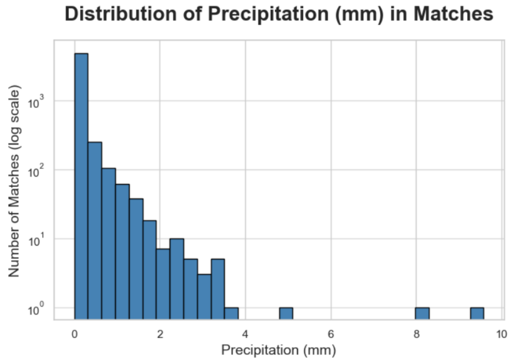
4.1.1 The Impact of Weather Conditions on Match Outcomes

To analyse match outcomes, we classified the match results into three categories: home win, draw, and away win. The general approach initially relied on descriptive visualizations, such as histograms and pie charts, to provide an intuitive overview of the distribution of results under different weather conditions. Subsequently, the chi-square test of independence was employed, as it is the standard method for testing whether two categorical variables are statistically associated.

Figure 4.1 shows both the number of matches played under rainy and non-rainy conditions and the distribution of precipitation values. The histogram highlights that matches without precipitation are substantially more frequent than those played with rain. For this reason, precipitation was treated as a binary variable, distinguishing between matches with no rain (0 mm) and matches with rain ($>$ 0 mm).



(a) Number of matches by precipitation category



(b) Distribution of precipitation values

Figure 4.1: Precipitation overview

As previously stated, the pie charts in Figure 4.2 show that 3,607 matches were played without rain, compared to 1,723 matches played with rain. In particular, the percentage distribution of match outcomes appears nearly identical across the two categories, suggesting that precipitation may not have a strong influence on match results.

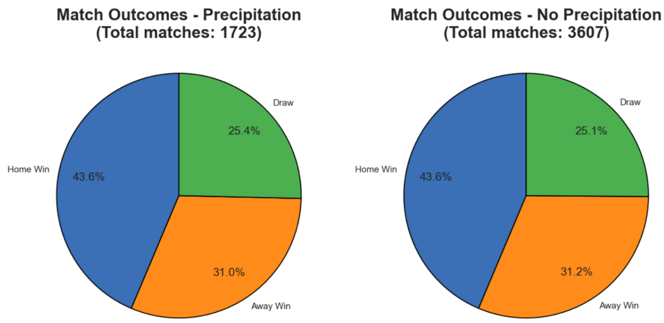


Figure 4.2: Distribution of match outcomes by precipitation

To formally test this observation and understand whether precipitation actually had an impact on match results, a chi-square test of independence was performed. Table 4.1 reports the observed frequencies for each category, while Table 4.2 shows the expected frequencies under the assumption of independence between precipitation and match outcome.

Precipitation	**Away Win**	**Draw**	**Home Win**
No	1127	906	1574
Yes	534	437	752

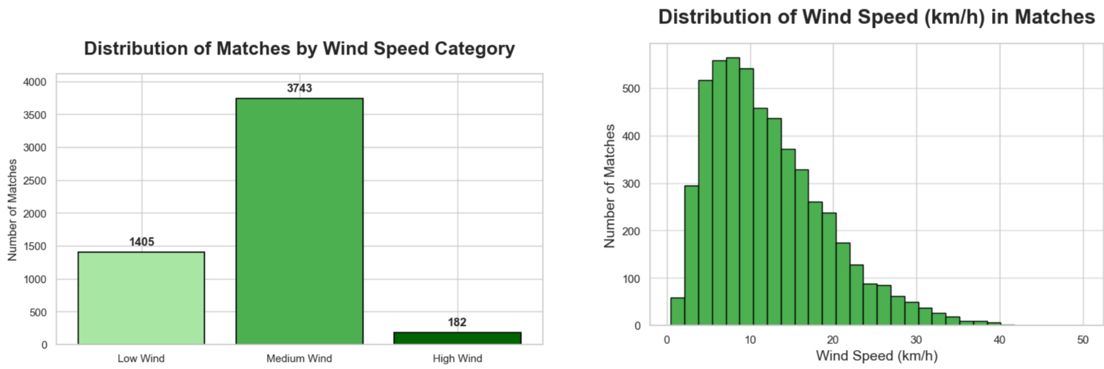
Table 4.1: Contingency table between precipitation and match outcomes

Precipitation	Away Win	Draw	Home Win
No	1124.06	908.86	1574.09
Yes	536.94	434.14	751.91

Table 4.2: Expected frequencies assuming independence between precipitation and match outcomes

The analysis yielded a chi-square statistic of 0.05 with 2 degrees of freedom and a p-value of 0.975, which is well above the conventional 0.05 significance level. Thus, the null hypothesis of independence cannot be rejected, indicating that, within the considered dataset, precipitation does not have a statistically significant effect on match outcomes. However, the analysis confirms a well-known pattern: playing at home is associated with a higher probability of winning, regardless of weather conditions. This suggests that other factors may have a more prominent role than rain in determining match results.

Then, we categorized wind conditions into three levels: low, medium, and high. Figure 4.3 presents both the number of matches per wind category and the distribution of wind speed values. Matches played under medium wind conditions are clearly the most frequent.



(a) Number of matches by wind speed category

(b) Distribution of wind speed values

Figure 4.3: Wind speed overview

The pie charts in Figure 4.4 suggest some variation in outcome distributions across wind categories, especially between low and high speeds. To assess whether these differences are statistically significant, we conducted a chi-square test of independence.

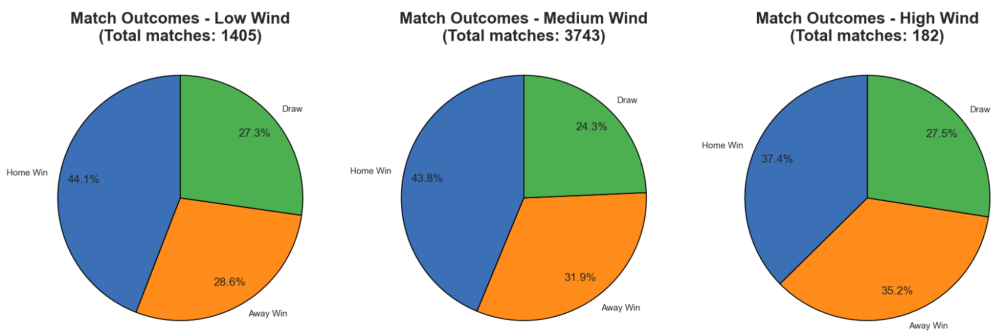


Figure 4.4: Distribution of match outcomes by wind speed

Observed and expected frequencies for each wind category, evaluated by the test, are reported in Tables 4.3 and 4.4, respectively. The analysis yielded a chi-square statistic of 10.95 with 4 degrees of freedom and a p-value of 0.037, which is below the 0.05 significance threshold. This indicates a statistically significant association between wind intensity and match outcome, allowing us to reject the null hypothesis of independence.

Wind Category	**Away Win**	**Draw**	**Home Win**
Low	402	383	620
Medium	1195	910	1638
High	64	50	68

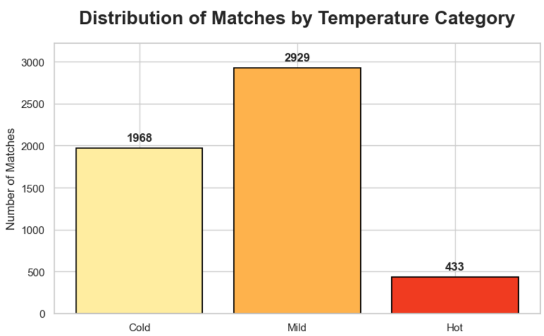
Table 4.3: Contingency table between wind speed and match outcomes

Wind Category	**Away Win**	**Draw**	**Home Win**
Low	437.84	354.02	613.14
Medium	1166.44	943.12	1633.44
High	56.72	45.86	79.42

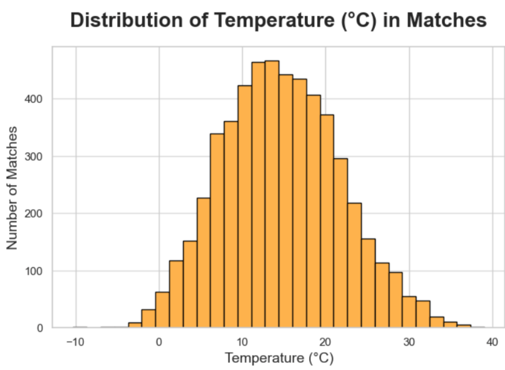
Table 4.4: Expected frequencies assuming independence between wind speed and match outcomes

This result suggests that wind can have a tangible impact on team performance. A plausible explanation is that strong winds affect ball trajectory and control, increasing uncertainty in passing and shooting despite high technical skill levels of players and modern pitch conditions. As a consequence, match dynamics and outcomes may be altered under windy conditions.

The last category analysed concerned temperature, which was divided into cold, mild, and hot. Figure 4.5 shows both the number of matches per category and the distribution of temperature values.



(a) Number of matches by temperature category



(b) Distribution of temperatute values

Figure 4.5: Temperature overview

Furthermore, as observed for precipitation and shown in Figure 4.6, the outcome distributions across temperature categories appear very similar, suggesting that temperature may also not have a significant impact on match outcomes.

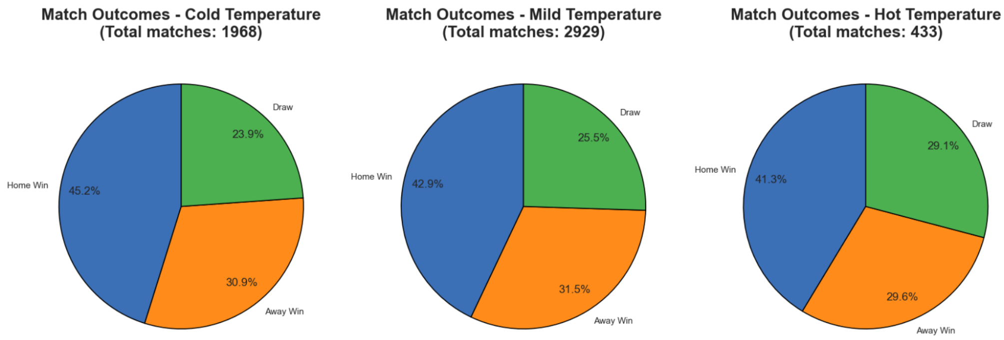


Figure 4.6: Distribution of match outcomes by temperature

To confirm this visual impression, we repeated the same approach adopted in the previous analysis, examining the statistically significant effect of temperature on match outcomes through a chi-square test. Tables 4.5 and 4.6 report the observed and expected frequencies, respectively. The analysis yielded a chi-square statistic of 6.50 with 4 degrees of freedom and a p-value of 0.164, exceeding the 0.05 significance level.

Temperature	**Away Win**	**Draw**	**Home Win**
Cold	609	470	889
Mild	924	747	1258
Hot	128	126	179

Table 4.5: Contingency table between temperature and match outcomes

Temperature	**Away Win**	**Draw**	**Home Win**
Cold	613.29	495.88	858.83
Mild	912.77	738.02	1278.21
Hot	134.94	109.10	188.96

Table 4.6: Expected frequencies assuming independence between temperature and match outcomes

Consequently, the null hypothesis of independence cannot be rejected, indicating that temperature does not have a statistically significant effect on match outcomes in the analysed dataset. This result is consistent with the findings for precipitation and suggests that match outcomes are largely robust to temperature variations.

4.1.2 The Impact of Weather Conditions on Match Statistics

After examining the impact of meteorological conditions on match outcomes, the analysis shifted to match statistics in order to capture more subtle effects of weather on game intensity and playing style. Initially, boxplots were used to visually compare statistics across different weather conditions, and for clarity, only a subset of these boxplots is reported. While the visual differences appeared limited, some variations were nonetheless observable. Additionally, to systematically determine the statistics most affected by weather conditions, an initial attempt considered differences in medians, as they are less sensitive to outliers. However, this approach was found to be inappropriate, given that game statistics are expressed in different scales and units of measurement. To overcome this limitation, Cohen's d was employed as a standardized, dimensionless measure of effect size. By expressing differences in terms of standard deviations, this metric enabled direct comparisons across diverse variables. Consequently, it was possible to rank and identify the game statistics most affected by weather conditions. Following Cohen's established conventions, effect sizes were interpreted as small ($d = 0.2$), medium ($d = 0.5$), or large ($d = 0.8$)

Figure 4.7 shows boxplots comparing match statistics under rainy and non-rainy conditions. For most variables, differences are present but visually minimal.

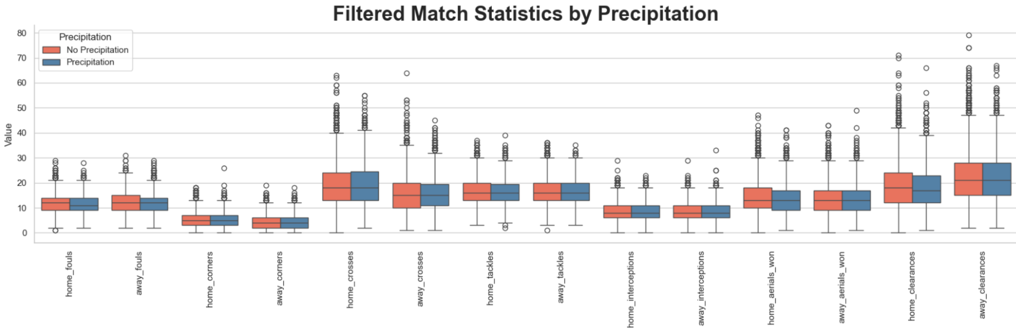


Figure 4.7: Boxplots for match statistics by precipitation

To verify these differences, we computed Cohen's d for each statistic. For greater readability, Table 4.7 reports the top 15 statistics with the highest absolute effect sizes, while Figure 4.8 provides a visual summary through a diverging chart.

Metric	**Cohen's d**	**Absolute Cohen's d**
Home passes attempted	0.095	0.095
Home touches	0.094	0.094
Home xG	0.093	0.093
Home passes completed	0.090	0.090
Away goals	0.085	0.085
Home goals	0.082	0.082
Away possession	-0.079	0.079
Home possession	0.079	0.079
Away interceptions	0.071	0.071
Home shots total	0.069	0.069
Home clearances	-0.059	0.059
Home offsides	-0.054	0.054
Home red cards	-0.054	0.054
Home corners	0.054	0.054
Away saves faced	0.053	0.053

Table 4.7: Top 15 match statistics most affected by precipitation (Cohen's d)

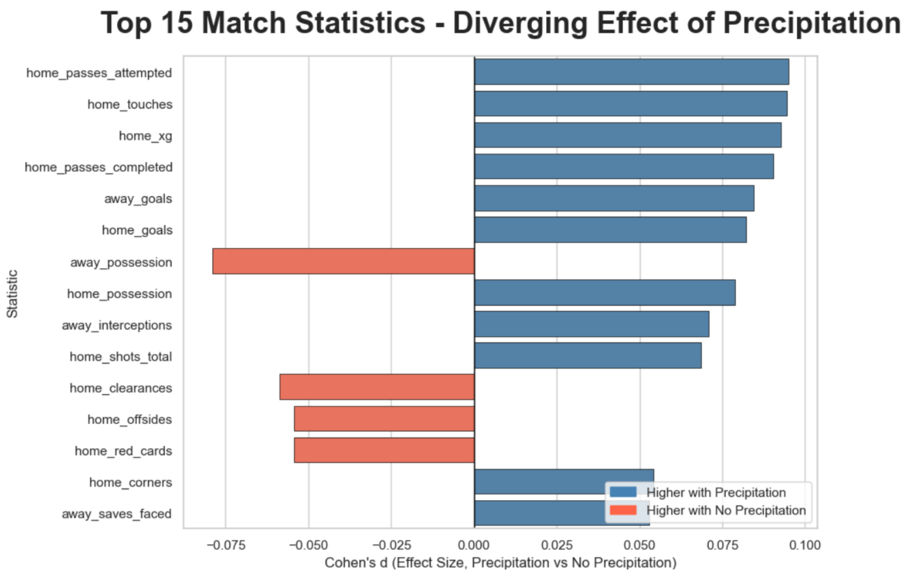


Figure 4.8: Diverging chart for top 15 match statistics most affected by precipitation

The results indicate that rain is associated with slight increases in home team metrics, such as passes attempted and completed, touches, and expected goals. Conversely, small decreases are observed in away team possession and some defensive actions, such as clearances. However, all effect sizes are less than 0.2, indicating that these differences are very small and practically negligible.

The same analytical framework was applied to wind conditions. Boxplots in Figure 4.9 suggest more noticeable differences across wind categories, especially through a stepped trend.

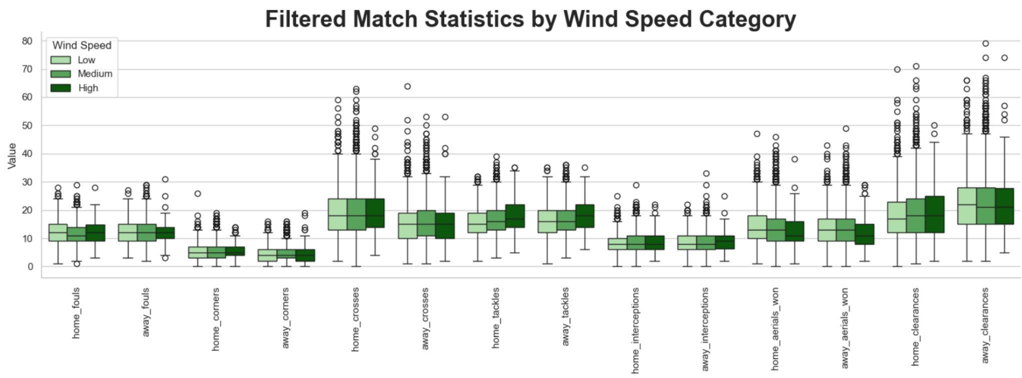


Figure 4.9: Boxplots for match statistics by wind speed

Subsequently, Cohen's d was calculated for each statistic by comparing matches played under high and low wind conditions to maximize the possible contrast. The variables with the largest effect sizes were then selected and reported in Table 4.8, as well as visualized in Figure 4.10.

Metric	Cohen's d	Absolute Cohen's d
Home tackles	0.452	0.452
Away tackles	0.346	0.346
Home aerials won	-0.272	0.272
Away aerials won	-0.255	0.255
Away goals	0.229	0.229
Away xG	0.228	0.228
Away shots total	0.219	0.219
Away long balls	-0.216	0.216
Home saves faced	0.186	0.186
Away shots on target	0.186	0.186
Home interceptions	0.164	0.164
Away touches	0.160	0.160
Away interceptions	0.140	0.140
Home touches	0.125	0.125
Home goals	-0.114	0.114

Table 4.8: Top 15 match statistics most affected by wind speed (Cohen's d)

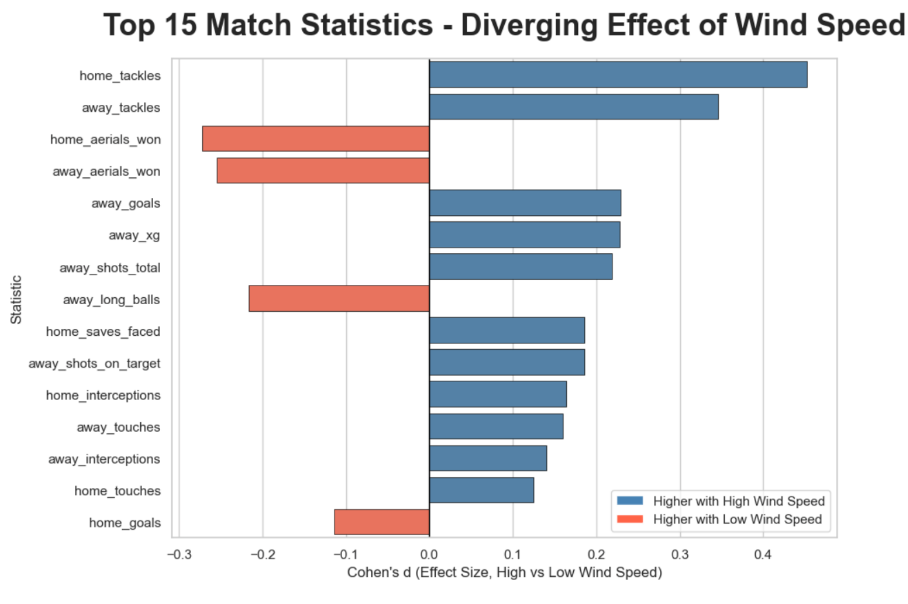


Figure 4.10: Diverging chart for top 15 match statistics most affected by wind speed

In this case, the analysis shows that the effects of wind are more pronounced than those observed for rain, with some effect sizes approaching a medium magnitude. Variables related to physical duels and defensive intensity, such as tackles and aerial duels, are particularly affected. Additionally, a slight increase in some away team offensive metrics, including shots, expected goals, and goals, is also observed. Conversely, more technical statistics, such as long balls and aerial duels, appear to be negatively affected by the presence of wind. Overall, these findings indicate that wind has a stronger influence on match statistics than rain, contributing to a more fragmented and less fluid style of play, especially during adversarial weather conditions.

Finally, the impact of temperature on match statistics was analysed using the same methodology. Boxplots comparing cold, mild, and hot conditions are shown in Figure 4.11, which revealed generally small differences.

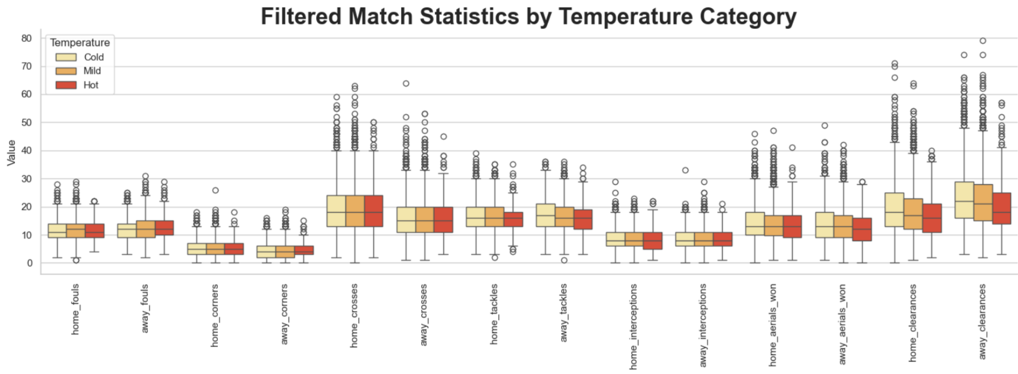


Figure 4.11: Boxplots for match statistics by temperature

Then, the effect of temperature was quantified by calculating Cohen's d for each variable, contrasting matches played in hot and cold conditions. The most affected statistics are summarized in Table 4.9 and displayed through the diverging chart in Figure 4.12.

Metric	Cohen's d	Absolute Cohen's d
Away clearances	-0.329	0.329
Home touches	-0.320	0.320
Home clearances	-0.302	0.302
Home passes attempted	-0.255	0.255
Home red cards	0.243	0.243
Home goal kicks	0.238	0.238
Away touches	-0.220	0.220
Away aerials won	-0.216	0.216
Away tackles	-0.210	0.210
Home passes completed	-0.205	0.205
Away goal kicks	0.194	0.194
Home aerials won	-0.190	0.190
Away yellow cards	0.188	0.188
Home tackles	-0.182	0.182
Home offsides	0.178	0.178

Table 4.9: Top 15 match statistics most affected by temperature (Cohen's d)

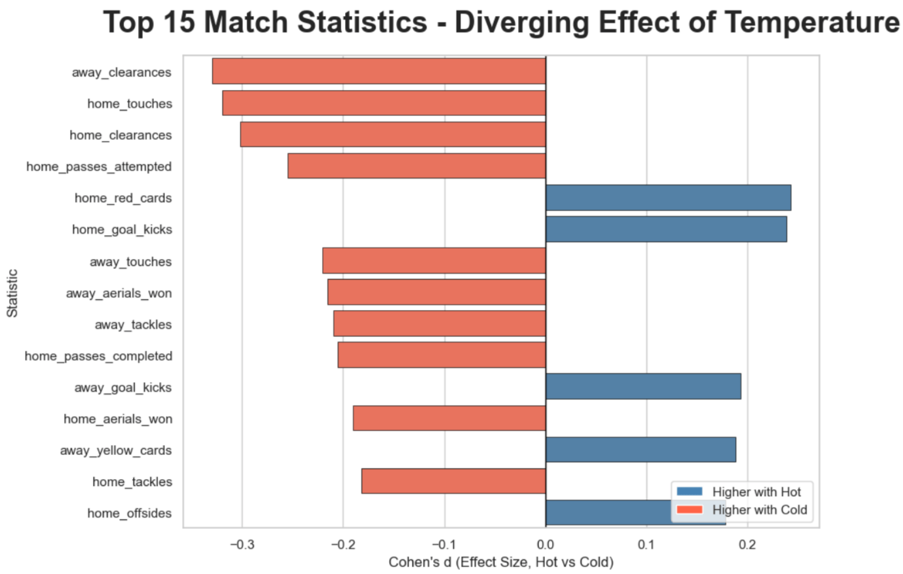


Figure 4.12: Diverging chart for top 15 match statistics most affected by temperature

The results suggest that temperature mainly influences match intensity and continuity. In colder conditions, physically demanding actions such as passes, tackles, touches, and aerial duels tend to occur more frequently, resulting in a more continuous and intense style of play. In contrast, matches played in hotter conditions appear slower and more interrupted, with slight increases in stoppages and disciplinary situations, including offsides.

4.2 Do Seasonal Effects influence Match Outcomes and Match Statistics?

While meteorological conditions offer insight into immediate environmental factors affecting football matches, seasonal patterns provide a broader perspective on how performance varies throughout the competitive year. In fact, seasonal effects may persist even after controlling for weather, since factors beyond meteorology, such as accumulated fatigue or strategic periodization, play significant roles in shaping performance over time.

Therefore, this research question investigates whether there are differences across seasons (spring, summer, autumn, and winter) in both match outcomes and team performance statistics. To address this, the analysis is also structured into two complementary parts. First, the study assesses how seasonality shapes match outcomes and explores whether the distribution of results, in terms of home wins, draws, and away wins, varies significantly across seasons. Second, the impact of seasonality on performance metrics examines match statistics to understand how intensity and playing style evolve across seasons.

4.2.1 The Impact of Seasonal Effects on Match Outcomes

Before examining seasonal effects on match outcomes, it is useful to assess the distribution of matches across different seasons. In the following analysis, to align with the standard football schedule, the data is organized from summer to spring, following the progression of the competitive year from kickoff to conclusion. Figure 4.13 reveals significant variation in match frequency, with spring recording the highest volume of 1,752 matches, whereas summer shows notably fewer at 482. This uneven distribution reflects the typical European football calendar, where matches are concentrated during the standard competitive season, ranging from August to May, with summer representing a short period that may include season opening matches, late cup competitions, or pre-season games.

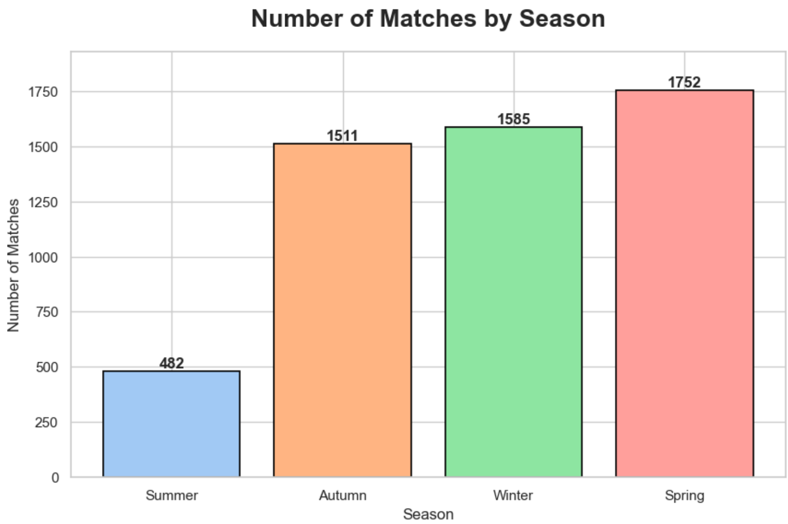


Figure 4.13: Seasonality overview

Consequently, to determine whether the season influences match outcomes beyond this imbalance, a chi-square test of independence was conducted on the contingency table of match outcomes (home win, draw, away win) across the four seasons, as shown in Table 4.10, since it accounts for sample size differences when assessing statistical significance.

Season	Away Win	Draw	Home Win
Summer	158	133	191
Autumn	444	376	691
Winter	503	399	683
Spring	556	435	761

Table 4.10: Contingency table between season and match outcomes

To assess whether these differences reflect genuine seasonal effects, the expected frequencies under the assumption of independence were computed, as shown in Table 4.11.

Season	Away Win	Draw	Home Win
Summer	150.21	121.45	210.34
Autumn	470.88	380.73	659.40
Winter	493.94	399.37	691.69
Spring	545.98	441.45	764.57

Table 4.11: Expected frequencies assuming independence between season and match outcomes

The analysis yielded a chi-square statistic of 6.96 with 6 degrees of freedom and a p-value of 0.325, which considerably exceeds the conventional significance level of 0.05 and fails to reject the null hypothesis of independence between season and match outcome. Therefore, this result shows no statistically significant evidence that the season during which a match is played systematically influences whether it results in a home win, draw, or away win.

Furthermore, examining the relative outcome distributions in Figure 4.14 supports these results. Across all seasons, home wins remain the most frequent outcome and although slight seasonal variations are observed (i.e., autumn exhibiting the highest home advantages at 45.7%, while summer only 39.6%), these differences still remain small in magnitude. More importantly, even summer, which includes fewer matches, displays outcome proportions comparable to those observed during the core football calendar. These stable distributions across seasons are consistent with the non-significant chi-square result and indicate that seasonal timing does not meaningfully influence match outcomes, which can be more reasonably driven by structural factors such as team strength, tactical approaches, and home advantage.

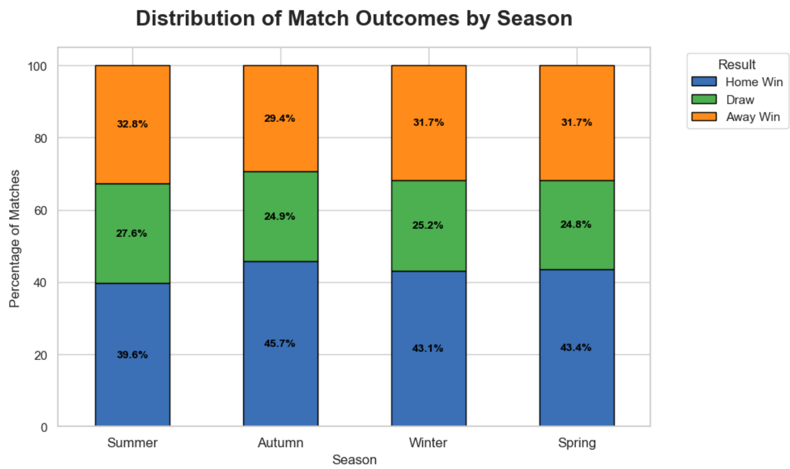


Figure 4.14: Distribution of match outcomes by season

4.2.2 The Impact of Seasonal Effects on Match Statistics

However, the previous findings do not exclude the possibility that playing style and match intensity vary across seasons. Even if the distribution of wins, draws, and losses remains stable, the underlying performance metrics, such as defensive actions or physical engagement, may differ systematically throughout the football year. Therefore, this section examines whether seasonal patterns affect how matches are played rather than who wins them.

To investigate whether and which performance metrics differ across seasons, a Kruskal-Wallis test was applied to each match statistic, comparing their distribution across the four seasons. In particular, the Kruskal-Wallis test was considered appropriate as it is non-parametric and does not require the assumptions of normality or homogeneity of variance, which is often not the case for football performance data.

Table 4.12 presents the top 15 statistics most affected by seasonality, ranked by their epsilon-squared value. All listed metrics show statistically significant differences across seasons ($p < 0.05$), with effect sizes, measured by epsilon-squared (ϵ^2), ranging from 0.0015 to 0.0163. Although effect sizes are small, they represent meaningful differences in professional football, where marginal performance variations matter. In particular, the strongest seasonal effects were observed in defensive metrics, like clearances, disciplinary records, such as yellow cards, and physical engagement indicators, including tackles and interceptions. These findings suggest that the intensity of play shows some seasonal variation. However, it is important to note that a significant Kruskal–Wallis test indicates the presence of seasonal variation but does not identify which seasons differ or the direction of these differences.

Metric	**H-statistic**	**p-value**	ϵ^2
Away clearances	89.68	<0.001	0.0163
Home clearances	49.78	<0.001	0.0088
Home yellow cards	37.51	<0.001	0.0065
Away tackles	31.25	<0.001	0.0053
Away yellow cards	25.51	<0.001	0.0042
Home tackles	23.70	<0.001	0.0039
Home interceptions	21.65	<0.001	0.0035
Away interceptions	21.28	<0.001	0.0034
Away goal kicks	18.88	<0.001	0.0030
Home aerials won	18.58	<0.001	0.0029
Home goal kicks	18.27	<0.001	0.0029
Away aerials won	16.78	<0.001	0.0026
Home touches	15.09	0.002	0.0023
Away xG	14.46	0.002	0.0022
Away throw-ins	11.00	0.012	0.0015

Table 4.12: Top 15 match metrics most affected by seasonality (Kruskal-Wallis test)

To clarify the direction of the observed seasonal effect, Table 4.13 reports the mean and standard deviation of the nine most season-sensitive performance metrics across the four seasons. Overall, the table shows that several defensive metrics, particularly clearances, increase from summer to winter, while tackling measures also rise but then decline in spring. Others, such as interceptions, peak earlier in the season and gradually decline. In contrast, metrics related to match flow, such as away goal kicks, show a slight decrease towards the final months, suggesting that later-season matches exhibit greater defensive intensity.

Metric	Summer	Autumn	Winter	Spring
Away clearances	19.63 ± 9.00	21.37 ± 9.38	23.52 ± 10.15	23.36 ± 10.55
Home clearances	17.37 ± 8.04	17.71 ± 7.95	19.71 ± 9.23	19.24 ± 9.30
Home yellow cards	1.81 ± 1.34	2.12 ± 1.44	1.98 ± 1.40	1.83 ± 1.34
Away tackles	16.42 ± 5.01	16.78 ± 5.30	17.28 ± 5.31	16.23 ± 5.17
Away yellow cards	2.33 ± 1.54	2.36 ± 1.51	2.21 ± 1.39	2.10 ± 1.42
Home tackles	15.96 ± 4.94	16.56 ± 5.21	16.82 ± 5.30	15.93 ± 5.03
Home interceptions	8.66 ± 3.60	8.49 ± 3.63	8.52 ± 3.61	8.03 ± 3.54
Away interceptions	8.96 ± 3.96	8.71 ± 3.55	8.76 ± 3.67	8.27 ± 3.47
Away goal kicks	8.26 ± 3.32	8.18 ± 3.51	7.85 ± 3.31	8.35 ± 3.44

Table 4.13: Summary statistics for top 9 metrics by season (mean ± standard deviation)

To further visualize seasonal patterns across performance metrics, Figure 4.15 presents a heatmap of normalized z-scores for the most season-sensitive metrics. By standardizing each metric relative to its seasonal mean, the heatmap emphasizes whether each season shows above-average values, indicated in red, or below-average values, shown in blue. The visualization reveals a clear concentration of positive deviations for defensive and physical engagement metrics in autumn and winter, while summer generally shows negative deviations, confirming that seasonality systematically shapes match dynamics.

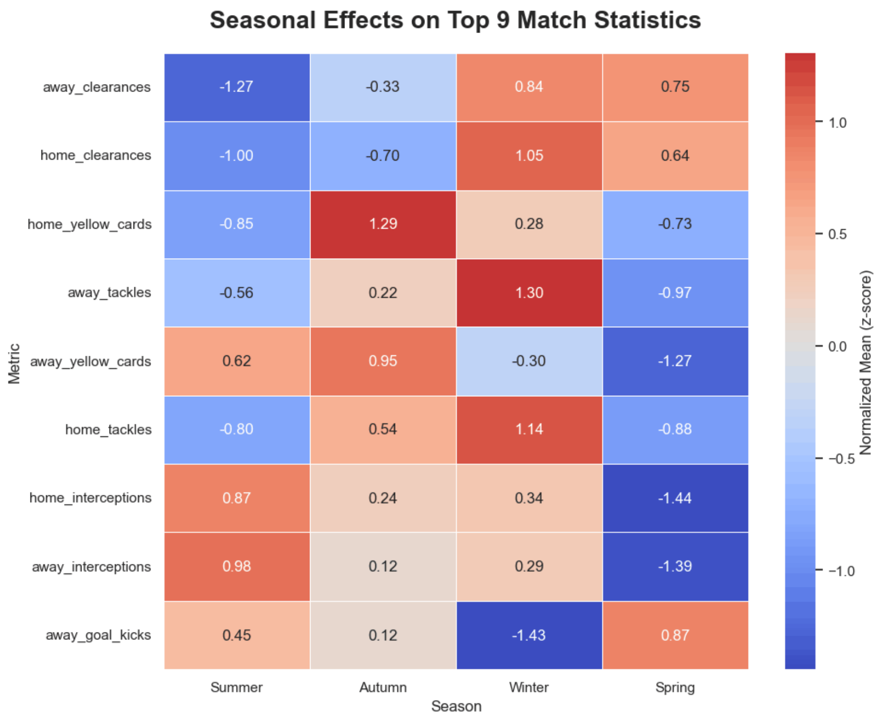


Figure 4.15: Seasonal effects on top 9 match metrics (normalized z-scores)

Then, to identify which specific seasonal pairs drive the observed differences, a post-hoc Dunn test with Bonferroni correction was applied to the most affected metrics. The Dunn test compares all pairs of groups, while the Bonferroni correction adjusts the p-values to control for the increased risk of false positives arising from multiple statistical tests. The resulting p-value matrices are visualized as heatmaps in Figure 4.16, where red cells indicate statistically significant differences ($p < 0.05$) and blue cells indicate non-significant comparisons.

The post-hoc analysis reveals that the seasonal effects are driven by differences between early-season (summer and autumn) and late-season periods (spring). In particular, for most defensive metrics like clearances, statistically significant differences emerge primarily between summer and the later seasons (winter and spring), while comparisons between neighboring seasons are more often non-significant. This fact indicates that seasonal variation in match performance evolves gradually rather than abruptly. Conversely, metrics related to match flow, such as away goal kicks, show significant contrasts between winter and the other seasons, reinforcing the inverse seasonal pattern observed in the descriptive analysis.

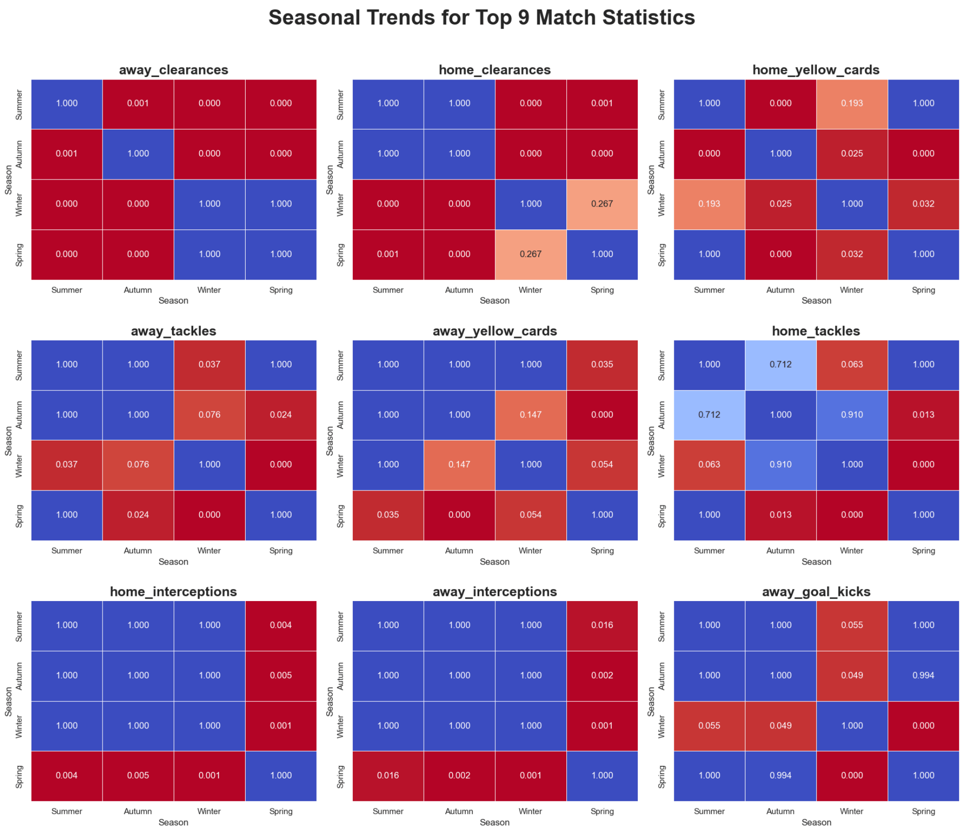


Figure 4.16: Pairwise seasonal comparisons for top 9 metrics (Dunn test with Bonferroni correction)

Finally, to better visualize the temporal progression of these patterns, Figure 4.17 presents line plots for the top 9 metrics, emphasizing the continuity and direction of seasonal changes. In particular, clearances rise steadily from summer to winter, reflecting increasing defensive intensity as the season progresses. Disciplinary indicators regarding yellow cards peak in autumn before declining through winter and spring, potentially suggesting both the physical intensity of the early season and a more rigorous enforcement of disciplinary directives. Other metrics, such as interceptions and some tackling measures, peak earlier in the year and gradually decline, suggesting that different components of defensive behaviour respond differently to seasonal dynamics. In contrast, match flow metrics, such as away goal kicks, reach their lowest values in winter before rising again in spring, reinforcing the inverse seasonal pattern observed in the descriptive statistics.

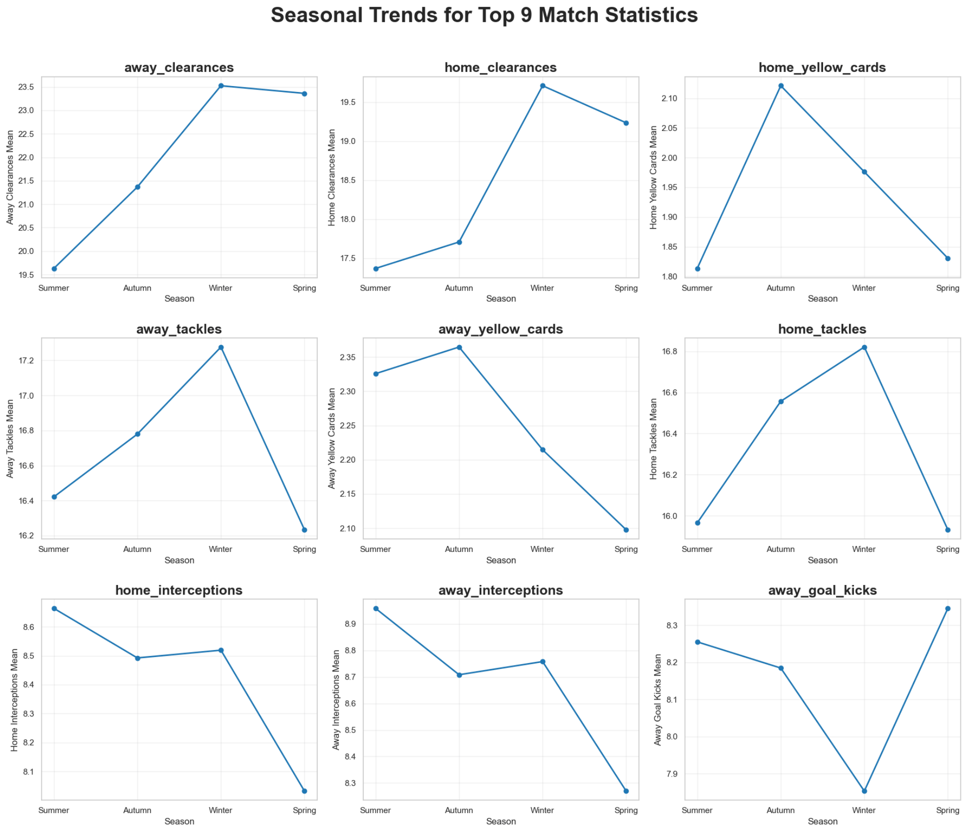


Figure 4.17: Seasonal trends for top 9 match metrics

Ultimately, these visualizations collectively demonstrate that seasonal effects on performance metrics follow systematic, interpretable patterns throughout the competitive year. While seasons do not significantly influence match outcomes, they affect how matches are played: defensive metrics like clearances increase toward winter, physical engagement shows varied patterns across the season, while match flow metrics slightly decrease.

5 Data Quality

Data quality analysis plays a fundamental role in data-driven projects. It especially determines whether the collected data are suitable for supporting reliable analysis and well-founded conclusions and provides a comprehensive framework for improvement. In this project, we conducted a data quality evaluation after the data acquisition, enrichment, database construction, and exploratory data analysis phases. This ex-post approach was intentionally adopted to validate the reliability of the analysis already performed and to verify whether potential data issues could meaningfully affect the interpretation of the results.

5.1 Quality Assessment

The main objective of data quality assessment is not only to detect errors but also to understand their nature and potential impact on the results. Given the relational nature of the database and its analytical purpose, we decided to primarily focus the assessment on two dimensions: completeness and consistency, which directly affect query reliability and result interpretability.

5.1.1 Completeness

Completeness evaluation is intended to verify whether the database contains all the information required to support meaningful analysis, as it indicates the coverage with which the observed phenomenon is represented in the database. For each entity stored in the database, completeness was considered at three different levels, namely table completeness, attribute completeness, and tuple completeness. Additionally, an overall completeness assessment was performed across all entities.

In general, the database exhibited a high level of completeness at all tiers. All expected entities and relationships were present, although very low percentages of null values were observed. These missing values were neither random nor explicit errors; rather, they could be explained by exceptional but legitimate situations in professional football competitions. In particular, one match (Montpellier vs. Saint-Étienne played on 16 March 2025) lacked detailed statistics because it was suspended before completion, while two other matches (Montpellier vs. Clermont Foot played on 29 November 2023 and Nantes vs. Lyon played on 7 April 2024) reported null attendance values because they were played behind closed doors. Overall, missing values affected less than 0.06% of the total records across all entities.

5.1.2 Consistency

Consistency checks were designed to verify whether the stored data comply with spatial, logical, and temporal constraints derived from the application domain, encompassing multiple complementary aspects. In particular, we evaluated three types of consistency:

- Spatial consistency verifies the geographical coherence between stadium coordinates and their corresponding countries, while also identifying redundant or overlapping stadiums.
- Logical consistency assesses whether the values stored in the database satisfy domain-specific rules and the internal relationships between attributes.
- Temporal consistency examines whether time-related attributes align with the temporal context in which the data is defined.

First, spatial consistency was assessed by exploiting PostGIS functionalities to analyse stadium coordinates. In particular, national borders were defined using country boundary geometries obtained from the Natural Earth dataset at 1:10m scale (`https://www.naturalearthdata.com/downloads/10m-cultural-vectors/10m-admin-0-countries/`), chosen for its global coverage. The corresponding table was added to the database following standard procedures to ensure the proper integration of the geometries. Then, all stadium points were spatially joined using a 1-kilometre outward buffer around national borders. This choice was motivated by the presence of three valid points: namely, the stadiums in Venice, Ajaccio, and Monaco, which are located directly on national borders or coastal areas and were not captured by a strict polygon intersection. Thus, this outward buffer ensured correct national assignment without introducing spurious inclusions, so that each stadium falls within its corresponding country, as shown in Figure 5.1.

Figure 5.1: Map of stadiums locations

For the analysis of overlapping stadiums, a 500-metre buffer was applied to each stadium location. This threshold was chosen to capture stadiums in close proximity that might represent duplicates, while avoiding false positives from legitimately nearby venues. The analysis identified twelve pairs of overlapping stadiums within the predefined buffer. Of these, ten pairs had identical coordinates, with a distance equal to 0 metres. The remaining two pairs, although located at a distance of less than 200 metres, were confirmed to refer to the same physical stadium, with the discrepancy attributable to minor variations in coordinate reporting. These duplicates do not represent serious data quality issues but rather reflect the dynamic nature of stadium nomenclature in professional football, where commercial partnerships frequently result in rebranding. For instance, the "Gtech Community Stadium" was formerly known as the "Brentford Community Stadium".

Second, logical consistency was based on the definition of a set of validation rules applied to the `Match_Statistics` table in order to evaluate some well-known football rules. Specifically, we verified that the number of completed passes does not exceed the number of attempted passes, that shots on target are not greater than total shots, and that disciplinary statistics regarding yellow cards and red cards are non-negative. Additionally, we evaluated whether the sum of ball possession percentages for the two teams equals one, and whether the relationship between saves made, shots on target, and goals conceded is respected when one of the teams scores zero goals. The analysis revealed no violations in passing accuracy, shooting statistics, or disciplinary data, indicating that these attributes are consistently recorded across all matches. However, eight matches were flagged by the possession consistency rule, as the sum of home possession and away possession slightly exceeded one, resulting in a total of 1.01. These deviations are minimal and are most likely due to rounding or approximation performed at the data source level, representing a limitation of the original reporting format rather than an actual data error. A more significant violation was detected for the rule relating to saves made, shots on target, and goals conceded. In one match, Union Berlin vs. Bochum played on 14 December 2024, the statistics indicate a 1-1 result in terms of saves and shots on target, while the final recorded score is 0-2. This inconsistency arises because the match was officially awarded to Bochum due to extraordinary circumstances, even though the statistics reflect the result achieved on the pitch. This case highlights a semantic inconsistency between the match outcome and performance statistics, rather than an error in data collection.

Third, temporal consistency evaluated the coherence between the date of each match and its corresponding competitive season. In the database, each competition is associated with a season identifier, such as 2023/2024. Thus, we verified that the year extracted from each match date belonged to one of the two calendar years defining the season. This validation did not reveal any inconsistencies, as all match dates correctly fall within the expected temporal range of their respective seasons, confirming the temporal alignment of all records.

Overall, the consistency assessment highlighted a limited number of anomalies, most of which are attributable to real-world exceptional circumstances or approximations inherited from the original source. As a result, the database can be considered largely coherent and suitable for analytical purposes, although there were some opportunities for targeted data quality improvements.

Before addressing data quality improvement, Table 5.1 summarizes the main findings regarding the quality assessment phase.

Dimension	Check	Issues	Notes
Completeness	Missing attendance values	2 matches	Matches played behind closed doors
	Missing match statistics	1 match	Match suspended before completion
Spatial consistency	Stadium-country coherence	0	All stadiums correctly assigned to their corresponding country
	Overlapping stadium locations	12 pairs	Duplicate records referring to the same physical stadium due to naming variations
Logical consistency	Completed vs. attempted passes	0	Completed passes never exceed attempted passes
	Shots on target vs. total shots	0	Shots on target always less than or equal to total shots
	Disciplinary statistics	0	Yellow and red cards always non-negative
	Ball possession sum	8 matches	Possession sums slightly exceeded 1 due to rounding effects (max 1.01)
	Saves, shots on target, goals	1 match	Officially awarded match
Temporal consistency	Season-date alignment	0	All match dates within the expected season boundaries

Table 5.1: Summary of data quality assessment

5.2 Quality Improvement

Following data quality inspection, a set of targeted quality improvement strategies was designed to mitigate the impact of the identified issues. Given the limited number of anomalies and their strong connection to real-world events, we implemented all improvements through database views, ensuring data traceability and reproducibility, rather than modifying or overwriting the original data. This choice preserves the integrity of the overall database while allowing us to operate with a refined representation of the data as well.

The spatial consistency assessment revealed multiple stadium records referring to the same physical venue but associated with slightly different names and identifiers, as addressed through `view_overlapping_stadiums`. Instead of removing or merging records, a canonical view, called `view_stadiums_clean`, was applied to map each stadium to a canonical identifier. This canonical identifier serves as a unique reference for the stadium, and any renaming in other seasons is mapped to it, resolving spatial redundancy while preserving the original identifiers, which continue to be referenced by other entities.

At the match level, three types of exceptions were identified during the assessment: matches that were suspended before completion, matches that were officially awarded after being played, and matches played behind closed doors. To address these issues, we defined a quality-aware view, namely `view_matches_quality`, by introducing three boolean indicators: `is_suspended`, `is_awarded`, and `closed_doors`, which identify these different exceptional cases, respectively. Then, a specific view, called `view_matches_clean`, was used to exclude suspended or awarded matches, whose available information does not fully or coherently reflect on-field performance, while matches with missing attendance values were not excluded. Furthermore, the `view_match_statistics_quality` was defined to focus on the logical consistency of possession data. Instead of enforcing a strict equality constraint, we applied a tolerance threshold of 0.015 to account for rounding effects introduced by the data source. Each match is therefore classified as either consistent or inconsistent with respect to ball possession, distinguishing acceptable approximations from actual errors. Then, the `view_match_statistics_clean` was used to retain only those match statistics whose possession values fall within the defined tolerance threshold, although no matches in our database violated this constraint after applying the threshold.

Ultimately, the improvements introduced through these views confirm that data quality issues affect only a very small subset of records and do not alter the overall structure or distribution of the data. This approach allows analytical flexibility while preserving the original data, making the quality management process transparent and reversible. Therefore, the exploratory analysis and research questions addressed earlier in the project remain valid and do not require recomputation, as the detected anomalies are either rare or related to exceptional real-world events.

6 Conclusions

This project provides an analysis of the relationship between environmental factors and football match performance by integrating match statistics from the top five European leagues with meteorological and seasonal data. Through systematic data collection, database design, and exploratory analysis, we addressed two research questions concerning the influence of weather conditions and seasonal effects on match outcomes and performance statistics.

Firstly, the analysis revealed that weather conditions have a limited impact on match outcomes. While precipitation and temperature showed no statistically significant association with the distribution of results, wind speed demonstrated a moderate effect. However, when examining match statistics, weather conditions displayed more complex influences on playing style and match intensity. In particular, wind had the strongest effect, specifically on physical duels, tackles, and aerial duels, with effect sizes approaching a medium magnitude. Temperature mainly affected match flow and continuity, with colder conditions associated with more physically demanding play. By contrast, precipitation showed minimal effects across all metrics examined.

Regarding seasonal effects, the analysis initially found that seasons do not significantly influence match outcomes, with the distribution of home wins, draws, and away wins remaining stable throughout the year. However, seasonality substantially affects how matches are played. Defensive metrics increased progressively from summer to winter, disciplinary indicators peaked in autumn before declining, and match flow metrics exhibited inverse patterns. These findings suggest that while the competitive balance remains consistent, the intensity and style of play evolve across the season, possibly reflecting accumulated fatigue, tactical periodization, or other factors beyond immediate weather conditions.

From a data management perspective, this project demonstrates the value of integrating structured match data with external information through a relational database design. The modular, pipeline-based architecture ensured maintainability and reproducibility, while the systematic data quality assessment confirmed the reliability of the findings. Ultimately, the use of PostgreSQL with PostGIS enabled efficient spatial queries and supported comprehensive analytical workflows.

6.1 Future Developments

While this project provides valuable insights, several limitations suggest directions for future work.

The current dataset, although comprehensive, only covers three seasons (2022/2023, 2023/2024, 2024/2025) of the top five European leagues. While sufficient for exploratory analysis, this limited period may not capture long-term patterns, such as climate trends or variations in team strategies. Therefore, future research could expand the dataset along two main dimensions: temporally, by including a larger number of seasons, and geographically, by incorporating leagues from different continents. Expanding the analysis to regions characterized by distinct meteorological conditions, together with a broader temporal window, would significantly enhance the robustness and generalizability of the findings. Moreover, the present study focuses exclusively on meteorological and seasonal patterns, whereas modeling time-dependent effects, such as the time of day at which matches are played, which is already recorded in the database but not explicitly analysed in this study, could represent a promising direction for future work.

Moreover, the data collection process required approximately eight hours to scrape 5,330 matches, posing a scalability challenge. To address this issue, future studies could develop an optimized and high-speed automated data acquisition system. This would involve implementing parallel processing and scheduled incremental updates to continuously expand the database at regular intervals. This approach would significantly reduce the computational time required for large-scale updates while ensuring the database remains current with the latest match performances.

The exploratory analysis used primarily traditional statistical methods. Future research could incorporate machine learning techniques, which are increasingly prominent in sports analytics. For example, supervised learning algorithms such as random forests or neural networks could predict match outcomes or performance metrics based on weather conditions, seasonal timing, and team characteristics, revealing complex interactions that uni-variate methods may overlook.

Despite these limitations, these results still offer practical value for clubs, analysts, and competition organizers. Understanding that weather rarely alters match outcomes but meaningfully shapes playing styles can help teams adjust training, tactical preparation, and team rotation under different environmental conditions. Likewise, recognizing that match intensity evolves across the season can support periodization planning and performance forecasting. More broadly, the integrated database and analytical framework developed in this project provide a scalable foundation for future decision-support tools in football analytics.